Recovery, Honest Grids, and Saddle Diagnostics

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-26

Abstract

This note explains the project lane that checks whether strong Raman suppression results survive stricter numerical validation, and what the remaining local geometry says about optimization. The main lesson is that recovery is not just re-running an optimizer: a result only counts if the pulse fits inside the simulated time window, energy drift is small, the same phase state can be found or revalidated, and the standard diagnostic images look physically sane. On that standard, several single-mode anchors survive, one simple-profile sweep is retired, and the high-performing branch remains saddle-like rather than minimum-like. The practical conclusion is that future optimizers should combine honest-grid validation, basis continuation, and explicit negative-curvature handling.

Claim Status

Status: established with scoped limits
The honest-grid recovery evidence is strong for the single-mode anchors shown in this note. The saddle evidence is strong for the canonical SMF-28 control ladder, but should not be overclaimed as a theorem for every fiber, power, or multimode setting. The retired sweep result is also important: not every visually impressive or deep result survives the boundary check.

1 Question and Thesis

The question is simple: when a saved optimization result looks good, how do we know it is a real Raman-suppression result and not a simulation artifact?

The thesis is that a result should pass three separate tests:

1. **physics test:** the output pulse stays inside the simulated time window instead of touching the edges;
2. **optimization test:** the reported objective still holds after revalidation or re-optimization on the repaired grid;
3. **geometry test:** if the optimizer stops, the local Hessian tells us whether the point is minimum-like or saddle-like.

Those tests change the story in a useful way. Some results become more credible. Some become weaker lower bounds. Some are retired. The saddle diagnostics also explain why first-order L-BFGS can stop at points that are useful but not true local minima.

Intuition Check

The easy mistake is to trust a small dB number by itself. A small number is not enough. The pulse also has to stay away from the time-window boundary, because otherwise the optimizer may be exploiting the numerical box instead of the fiber physics.

2 The Numerical Failure Mode

The forward model propagates a shaped input spectrum

$$u_\omega(0;\phi) = u_{\omega,0} e^{i\phi(\omega)}$$

through a nonlinear fiber model and measures the output Raman-band energy fraction,

$$J_{\text{phys}}(\phi) = \frac{\sum_{\omega_k \in \mathcal{B}_{\text{Raman}}} |u_{\omega_k}(L;\phi)|^2}{\sum_k |u_{\omega_k}(L;\phi)|^2}.$$

The reported value is

$$J_{\text{dB}}(\phi) = 10 \log_{10} J_{\text{phys}}(\phi).$$

This objective is only meaningful if the numerical grid is large enough. The time window has finite edges. If nonlinear broadening, dispersion, or an arbitrary group delay pushes the pulse into those edges, the FFT-periodic solver can wrap or contaminate the output. The recovery lane therefore measures an edge fraction,

$$\eta_{\text{edge}} = \frac{\int_{\text{edge}} |u(t,L)|^2 \, dt}{\int_{\text{all}} |u(t,L)|^2 \, dt},$$

where the edge region is the beginning and end of the simulated time window. The working acceptance rule is

$$\eta_{\text{edge}} < 10^{-3}.$$

Energy drift is checked separately:

$$\delta_E = \frac{|E_{\text{out}} - E_{\text{in}}|}{\max(E_{\text{in}}, \epsilon)}.$$

Low edge fraction checks boundary containment. Low energy drift checks that the propagator is not quietly losing or gaining too much energy.

Intuition Check

The edge check is like making sure a photograph did not crop off part of the object. If the pulse hits the edge of the time window, the optimizer may still report a nice number, but we no longer know whether that number came from real fiber dynamics.

3 Honest-Grid Recovery Method

The recovery workflow starts from an existing candidate phase, repairs the grid, and then decides whether the candidate survives.

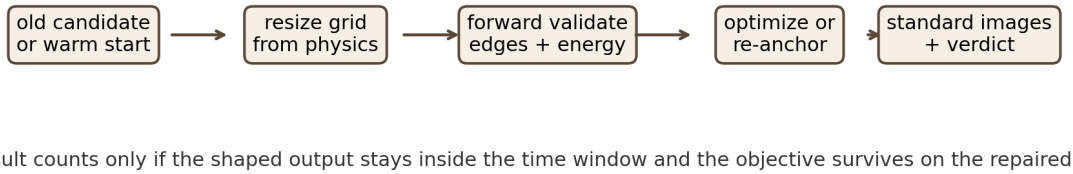


Figure 1: Honest-grid recovery loop. Old candidates are not accepted directly: the grid is resized from physical estimates, the field is forward-validated, optimization or re-anchoring is rerun when needed, and the final verdict is based on scalar diagnostics plus standard images.

The grid choice is not arbitrary. The first guess comes from the expected dispersive walk-off and nonlinear broadening. In words,

$$\text{window size} \approx \text{pulse width} + \text{dispersion walk-off} + \text{SPM-broadened safety margin}.$$

The estimate only proposes a starting grid. The actual gate is empirical: propagate the flat pulse and the relevant warm-started phase on the candidate grid, then double the time window if the output edge fraction fails.

For a saved phase ϕ_{old}, recovery also removes the two obvious phase gauges before judging the grid:

$$\phi_{\text{fixed}}(\omega) = \phi_{\text{old}}(\omega) - (a + b\omega).$$

The constant term is a global phase. The linear term is a time shift. Neither changes the Raman-band fraction physically, but both can make a saved phase look artificially wide in time if left in place.

Intuition Check

Subtracting $a + b\omega$ is not changing the real control in a meaningful way. It is just choosing a cleaner coordinate system. A constant phase and a rigid time shift are like moving the ruler, not changing the pulse physics.

4 Result Verdicts

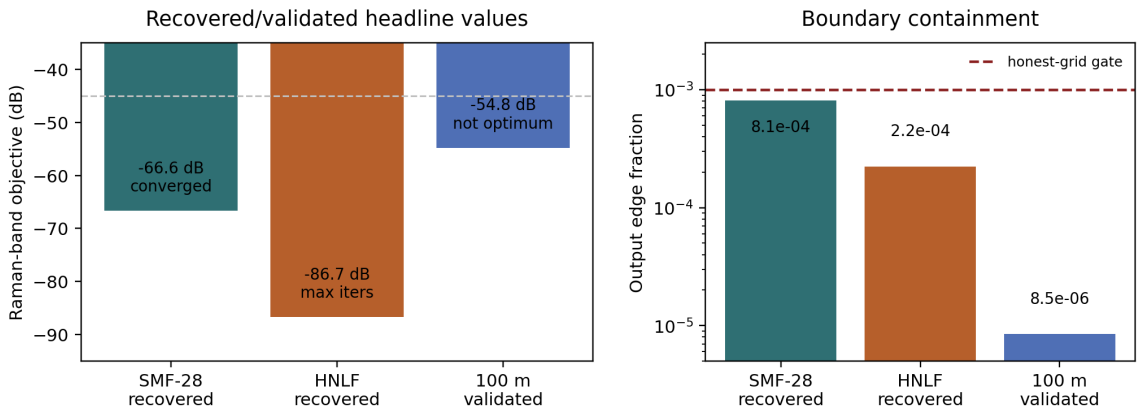


Figure 2: Recovered and validated anchors. The dashed edge-fraction line marks the 10^{-3} acceptance gate. The HNLF and 100 m entries are useful but should be described carefully because their stored optimizer flags did not certify convergence.

Case	Regime	Objective	Edge frac.	Verdict
Recovered SMF-28 anchor	$L = 2.0\,\mathrm{m}$, $P = 0.2\,\mathrm{W}$	$-66.61\,\mathrm{dB}$	8.10×10^{-4}	Recovered and converged on the repaired grid.
Recovered HNLF anchor	$L = 0.5\,\mathrm{m}$, $P = 0.01\,\mathrm{W}$	$-86.68\,\mathrm{dB}$	2.24×10^{-4}	Recovered, but the run stopped at the iteration cap. Treat as a strong achieved value, not a proved optimum.
Validated long-fiber lower bound	$L = 100\,\mathrm{m}$, $P = 0.05\,\mathrm{W}$	$-54.77\,\mathrm{dB}$	8.47×10^{-6}	Revalidated from the full saved state. Honest lower bound, not a convergence claim.
Retired simple-control sweep	$L = 2.0\,\mathrm{m}$, $P = 0.2\,\mathrm{W}$	best $-66.03\,\mathrm{dB}$	8.42×10^{-2}	Fails the boundary gate; do not use as a clean simplicity claim.

Intuition Check

The table has two columns that matter most: objective and edge fraction. A deep objective with a large edge fraction is not a clean win. A less dramatic result that passes the edge gate can be more scientifically useful.

5 Representative Result Pages

Each page pairs the phase diagnostic with the corresponding spectral-evolution heat map. The phase diagnostic says what input control was applied. The heat map says how the spectrum evolved inside the fiber.

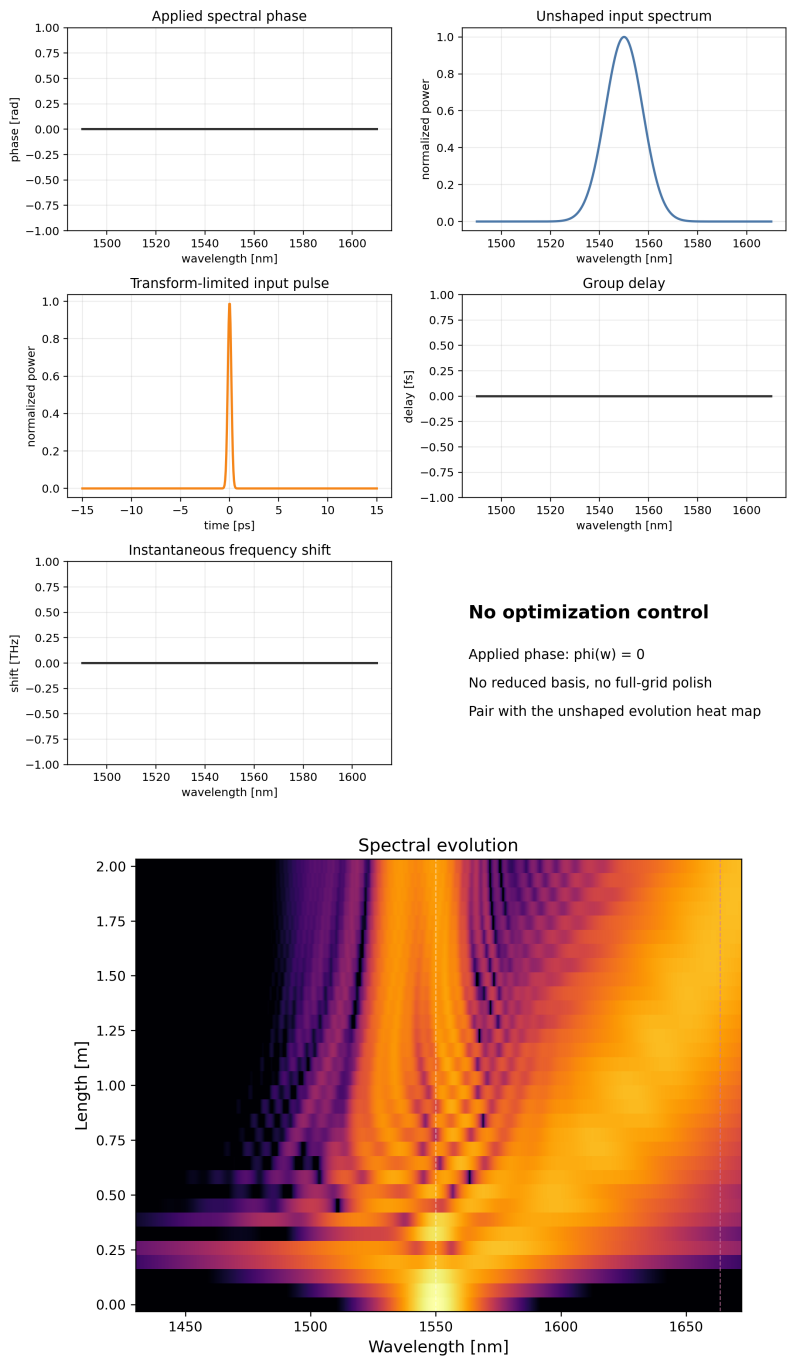


Figure 3: No-optimization control for the canonical SMF-28 setting. The flat phase is the reference group: it shows the spectral evolution before recovery or saddle diagnostics try to improve anything.

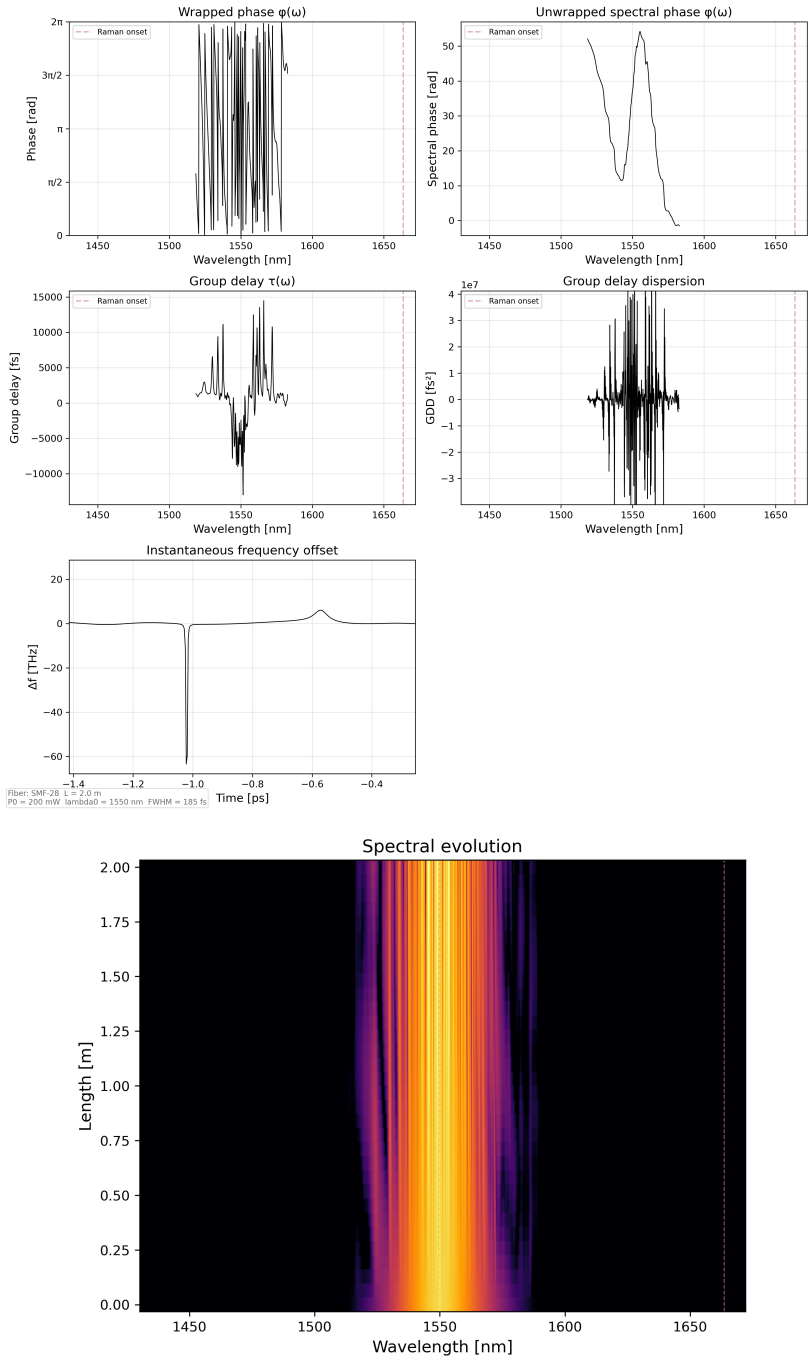
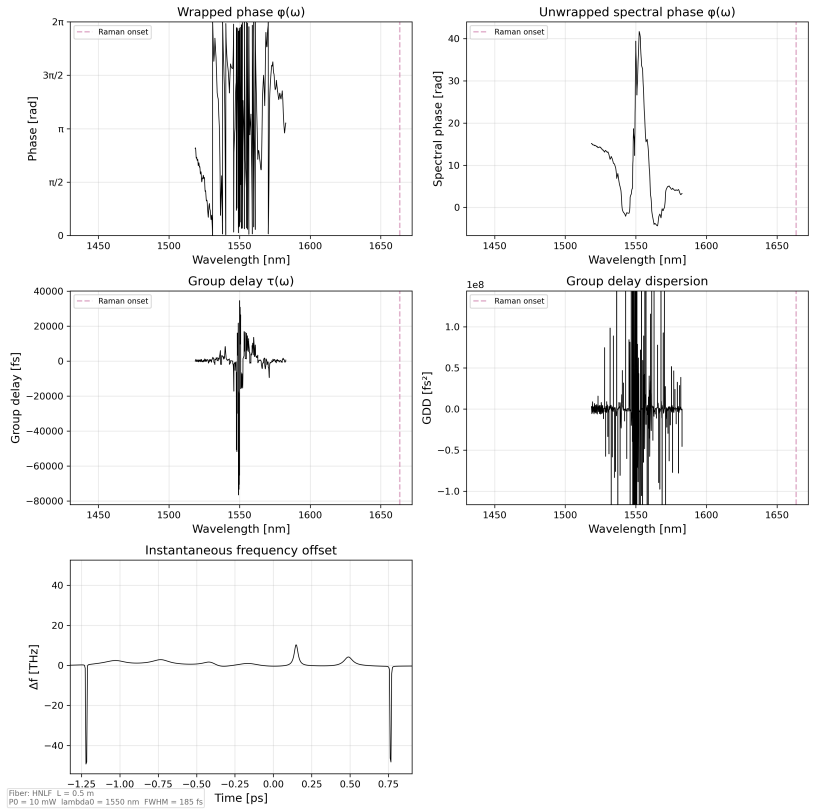


Figure 4: Recovered SMF-28 anchor. This is the cleanest recovered single-mode case in the note: it passes the boundary gate, converges, and keeps the standard phase-diagnostic plus heat-map evidence together.



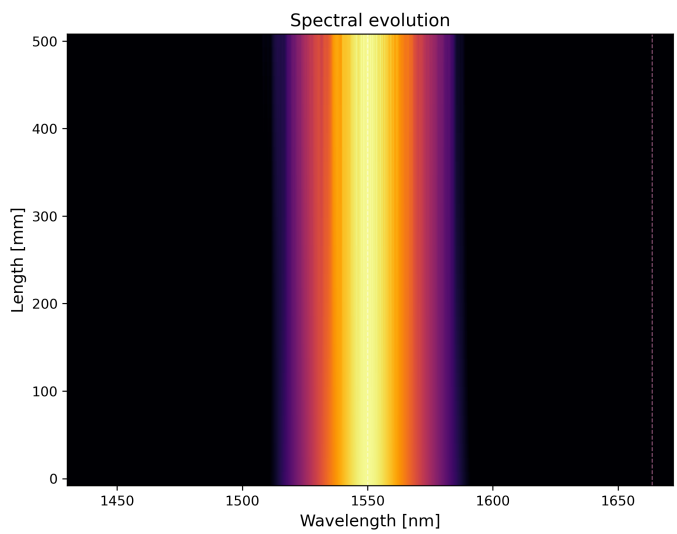


Figure 5: Recovered HNLF anchor. The value is very deep and the edge fraction is inside the gate, but the optimizer hit its iteration cap, so this page supports an achieved-value claim rather than an optimum claim.

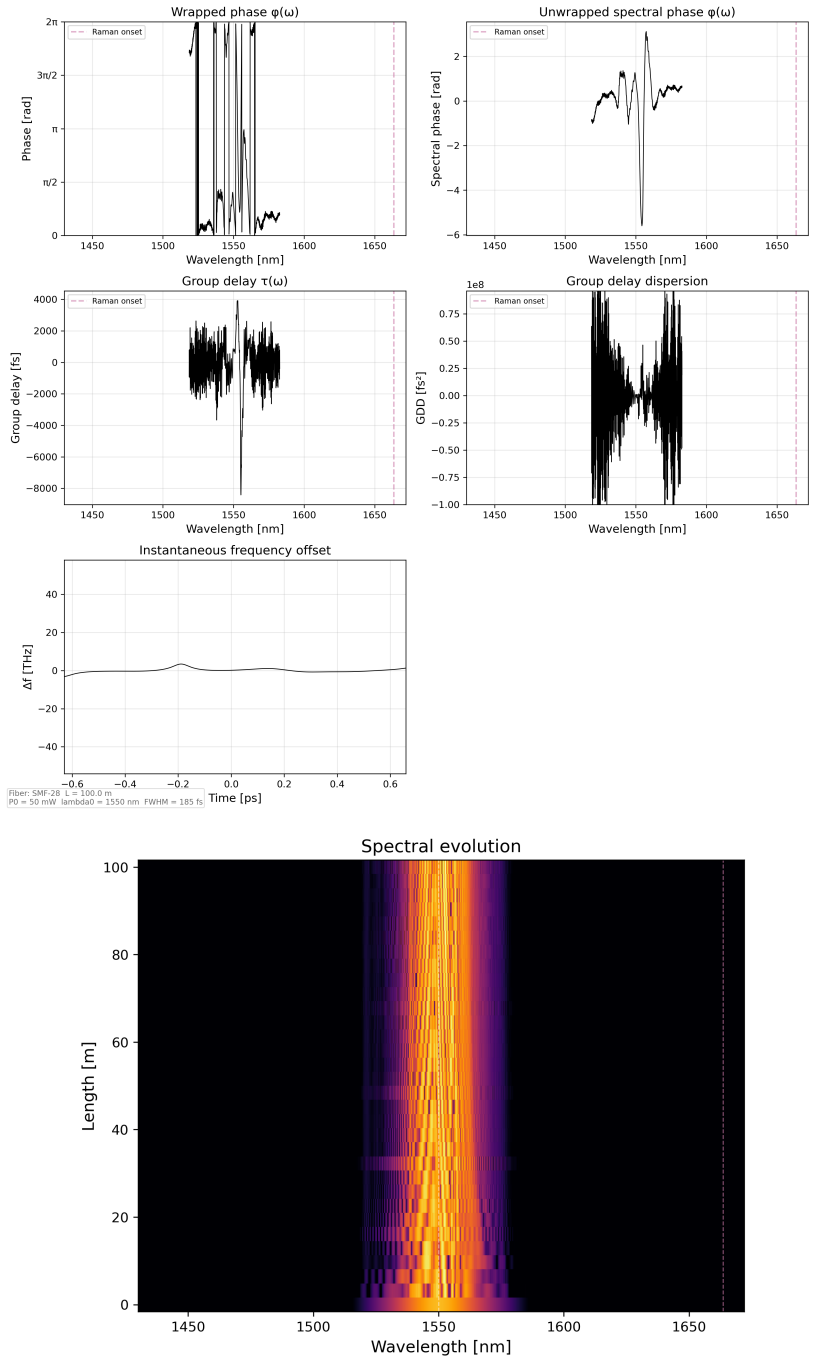


Figure 6: Validated long-fiber lower bound. The long propagation is numerically contained, but the stored run should be described as a validated achieved value, not as a solved global or local optimum.

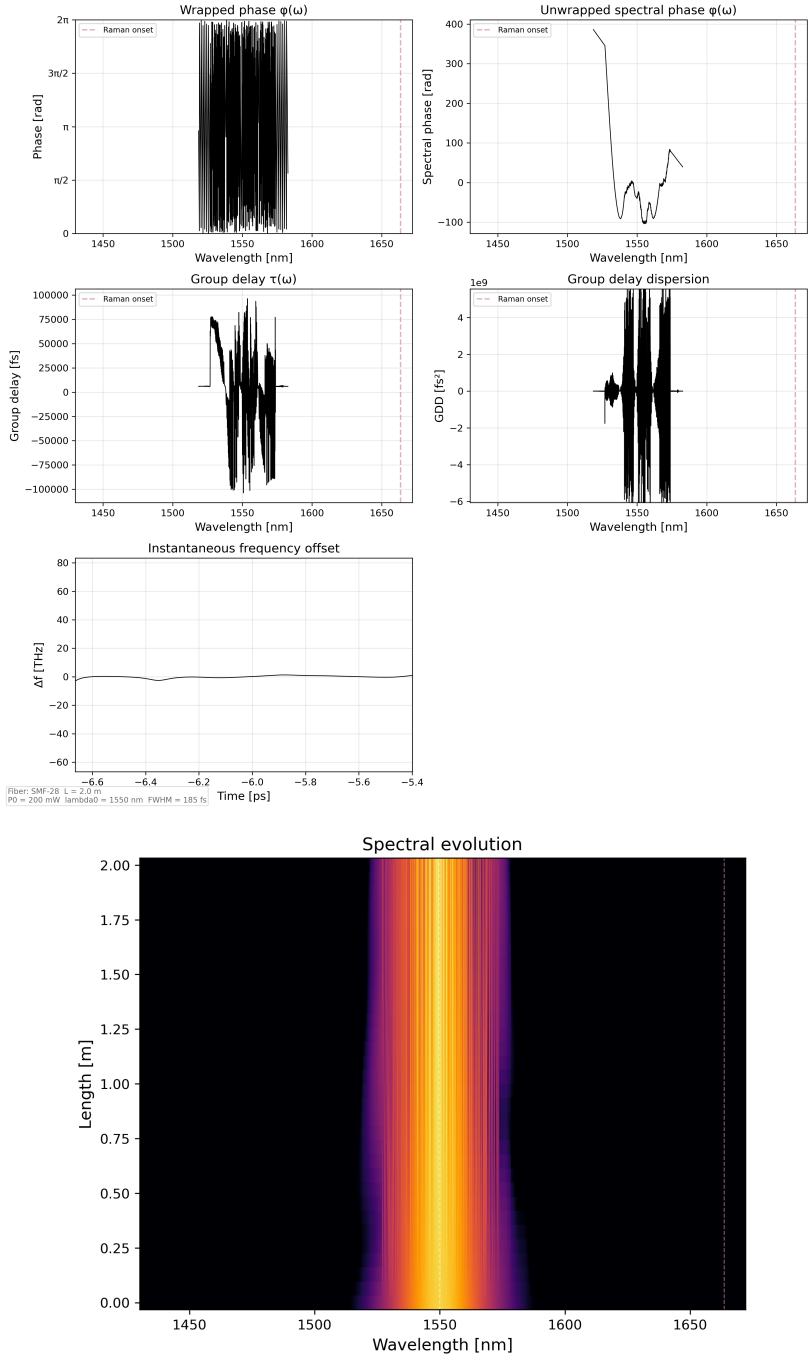


Figure 7: Best recovered point from the retired simple-control sweep. The objective is deep, but the edge fraction is far above the acceptance gate, so this image belongs in the note as a warning example rather than as a headline success.

6 The Retired Sweep

The simple-control sweep was originally attractive because it suggested a clean story: only a modest number of phase controls might be needed to reach nearly full-grid suppression. Recovery made that story weaker. Even on a much larger time window, the recovered sweep points kept too much energy near the boundary.

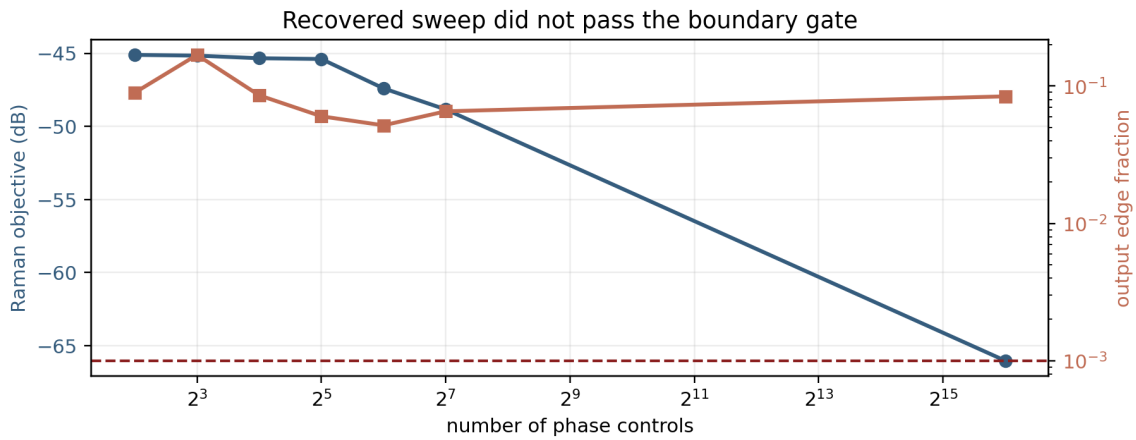


Figure 8: Recovered simple-control sweep. The best objective comes from the largest control space, but all recovered points remain well above the 10^{-3} edge-fraction gate. The right scientific move is to retire the clean simplicity claim until a boundary-aware rerun proves it.

Intuition Check

This is not a failed project result. It is a successful audit result. It tells us not to build a public claim on a run that may have learned the simulation window.

7 Saddle Diagnostics

Recovery tells us whether a point is numerically honest. It does not tell us what kind of local geometry the optimizer found. For that, the project studies the Hessian,

$$H(\phi) = \nabla_\phi^2 J_{\text{surf}}(\phi).$$

For a perturbation v, the local quadratic model is

$$J_{\text{surf}}(\phi + \epsilon v) \approx J_{\text{surf}}(\phi) + \epsilon \nabla J_{\text{surf}}(\phi)^T v + \frac{\epsilon^2}{2} v^T H(\phi) v.$$

At a stationary point, the gradient term is small. The sign of $v^T H v$ then decides whether that direction bends upward or downward.

- If all non-gauge eigenvalues are positive, the point is minimum-like.
- If some are positive and some are negative, the point is saddle-like.

- If the negative eigenvalues are small, first-order optimizers may stop there even though a downhill direction exists.

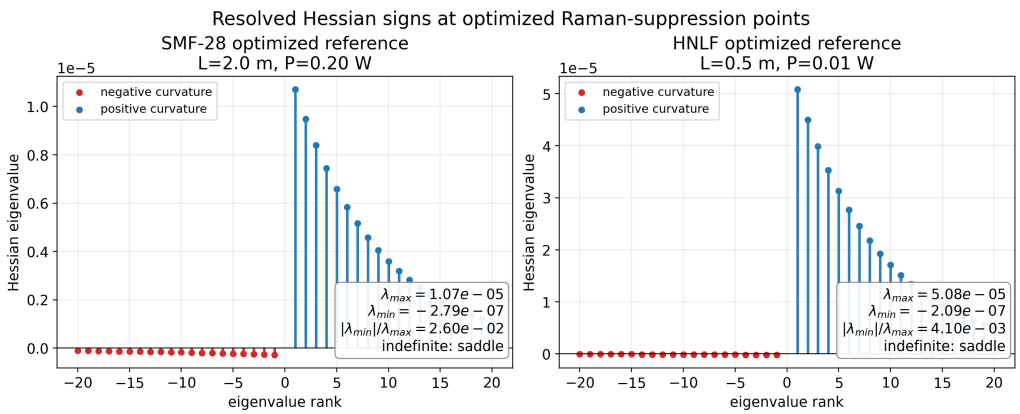


Figure 9: Resolved Hessian signs at optimized Raman-suppression points. The presence of both positive and negative curvature is the signature of a saddle, not a strict local minimum.

Intuition Check

The Hessian is the local "bowl test." A minimum is a bowl from every direction. A saddle is a mountain pass: if you move one way it goes up, but if you move another way it goes down.

8 Control-Space Ladder

The most useful saddle experiment restricts the phase to a small basis, then opens the basis gradually. Let $B \in \mathbb{R}^{N_t \times N_\phi}$ be a basis matrix and $c \in \mathbb{R}^{N_\phi}$ be the coefficient vector:

$$\phi = Bc.$$

The reduced objective is

$$f(c) = J_{\text{surf}}(Bc),$$

with gradient and Hessian

$$\nabla_c f = B^T \nabla_\phi J_{\text{surf}}, \qquad \nabla_c^2 f \approx B^T H_\phi B$$

plus the usual nonlinear terms from the dependence of the objective on ϕ. In practice, the reduced Hessian can be computed densely for small N_ϕ, which makes the local geometry much easier to classify.

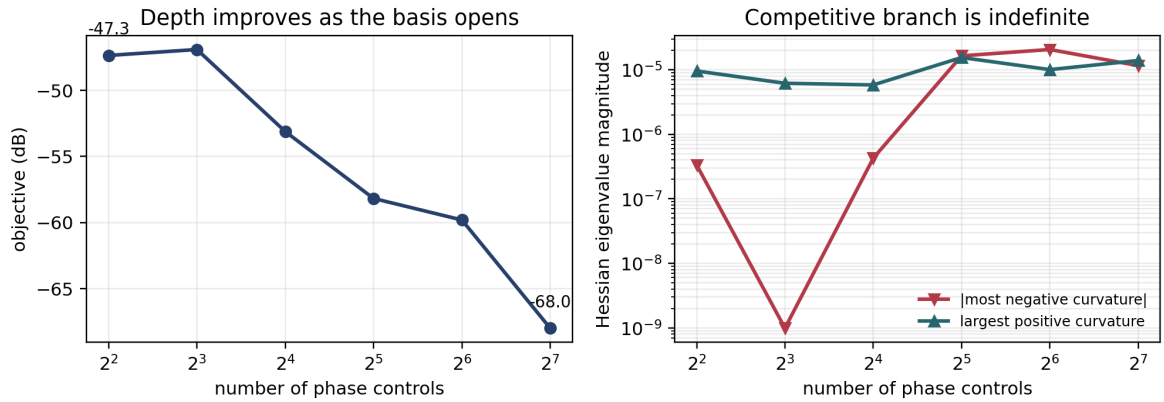


Figure 10: Control-space ladder at the canonical SMF-28 setting. The very small basis is minimum-like but shallow. The competitive branch appears once the basis opens, and that branch is already indefinite.

The important result is not just that saddles exist. The important result is where they appear. The minimum-like low-dimensional point is about 20 dB worse than the competitive branch. By the time the result is competitive, the Hessian is already indefinite.

Intuition Check

Restricting the basis is like forcing yourself to draw the phase mask with only a few smooth knobs. That can create a clean local minimum, but it may be too simple to get the best Raman suppression. Opening more knobs improves depth and also opens directions where the landscape bends downward.

9 Negative-Curvature Escape

If the smallest Hessian eigenvalue is negative, then its eigenvector points toward a local downhill direction in the quadratic model. A cheap escape strategy is:

$$\phi_{\text{new}} = \phi \pm \alpha v_{\min}, \qquad H v_{\min} = \lambda_{\min} v_{\min}, \qquad \lambda_{\min} < 0,$$

then restart the first-order optimizer.

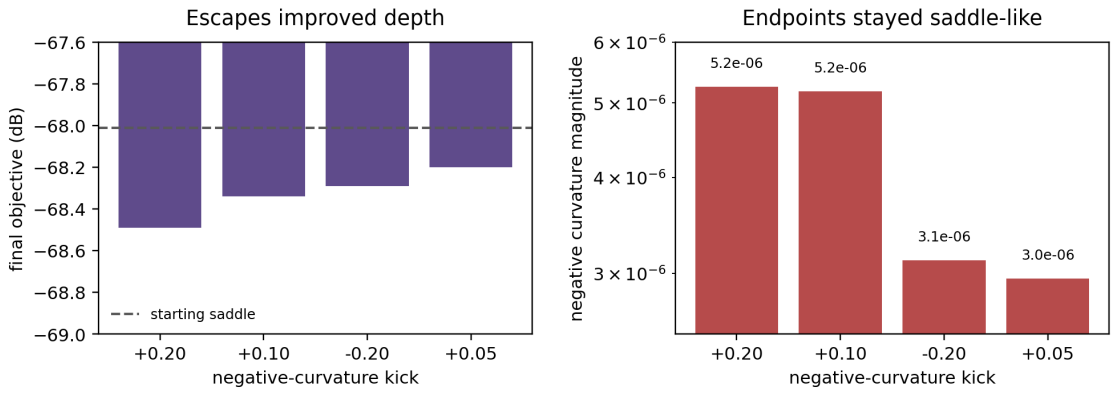


Figure 11: Negative-curvature escape from a competitive saddle. The escape steps improved objective depth by roughly 0.2–0.5 dB, but all tested endpoints remained saddle-like.

This is the practical optimization lesson: negative-curvature escape is useful, but it should be described honestly. In the tested setting it found better saddles, not clean minima.

Intuition Check

If L-BFGS stops at a mountain pass, a Hessian eigenvector can tell us which way the trail slopes down. But taking that trail does not guarantee we end in a valley. We may just reach another mountain pass at a lower elevation.

10 Implementation Surface

Component	Role
Recovery helpers	Grid sizing, gauge removal, forward validation, energy drift, boundary checks, and standard-image saving.
Sweep recovery driver	Re-runs the simple-control ladder on repaired grids and writes the retired-sweep summary.
Long-fiber validation reader	Pairs the scalar summary with the full saved state so the actual phase can be revalidated.
Saddle diagnostic driver	Builds reduced-basis Hessians, classifies curvature, and tests negative-curvature escape steps.
Standard-image helper	Writes the phase diagnostic and spectral-evolution heat maps used in the result pages above.

The human-facing note avoids internal run labels. For reproduction, the local README next to this PDF lists the relevant script families and result summaries without making them part of the scientific narrative.

11 What This Means For Future Work

Recovery first. A candidate should not be promoted until the no-shaping control, phase diagnostic, optimized heat map, unshaped heat map, edge fraction, and energy drift have all been checked.

Do not oversell achieved values. If a run passes the physics checks but does not converge, describe it as an achieved value or lower bound, not as an optimum.

Use saddle information operationally. The Hessian should guide restarts and continuation, not just decorate a report.

Best next optimizer direction. The evidence points toward reduced-basis continuation plus a globalized second-order method: Newton-CG, trust-region Newton, or cubic-regularized Newton with explicit negative-curvature handling [6, 7, 4].

12 Presentation Capsule

If this note becomes a validation slide, the clean message is:

- Recovery work is how the project separates real saved results from numerically fragile or retired claims.
- A candidate is not promoted from a pretty heat map alone; it needs the control, phase diagnostic, edge/energy checks, and honest-grid rerun.
- Saddle diagnostics explain why reruns, perturbations, and restarts are scientifically necessary rather than bookkeeping.
- The safest story is not "every old result was right." It is that the project now has a method for deciding which results survive.

13 Limitations

The saddle ladder is one operating point. The canonical SMF-28 result is important, but not a proof that every fiber and power behaves the same way.

The long-fiber result is not the long-fiber note. It appears here only because the recovery lane revalidated its saved state. A full long-fiber interpretation belongs in a separate note after that lane stabilizes.

No multimode claim is made here. Multimode recovery was not completed to the same standard in this evidence snapshot.

Images do not replace scalar gates. The phase profiles and heat maps are necessary for presentation, but the edge fraction and energy drift are what make the pictures trustworthy.

14 Reproduction Capsule

For an existing candidate phase, the recovery checklist is:

1. load the saved phase and its physical regime;
2. remove constant and linear phase gauge over the active spectral band;
3. choose a repaired time window from dispersion plus nonlinear broadening estimates;
4. forward-propagate the flat pulse and candidate phase;
5. require $\eta_{\text{edge}} < 10^{-3}$ before trusting the result;
6. rerun or re-anchor the optimization if the candidate survives;
7. save the standard image set and visually inspect it.

For saddle diagnostics, the checklist is:

1. choose a reduced basis $\phi = Bc$;
2. compute the dense reduced Hessian at the candidate;
3. classify the signs of the eigenvalues after ignoring gauge modes;
4. if $\lambda_{\min} < 0$, try a small step along $v_{\min}$;
5. restart optimization and re-check the final Hessian class.

15 Sources and References

References

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